📘 **AI/ML Internship – Day 10 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Introduction to OOP**

🔹 **What is OOP?**

Object-Oriented Programming (OOP) is a way of writing code using **objects and classes**.

👉 Instead of writing everything in functions, we organize code into:

- Classes
- Objects

🔹 **Why OOP?**

- Better structure
- Reusability
- Easy to manage large projects
- Used in real-world AI applications

📌 **Part 2: Classes and Objects**

🔹 **What is a Class?**

A class is a blueprint (template).

🔹 **What is an Object?**

An object is an instance of a class.

🔹 **Example**

```
class Student:
    name = ""
    marks = 0
```

```
s1 = Student()
s1.name = "John"
s1.marks = 85

print(s1.name, s1.marks)
```

📌 Part 3: Constructor (init)

🔹 What is init?

A constructor is used to initialize object values.

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

s1 = Student("Sara", 90)
print(s1.name, s1.marks)
```

📌 Part 4: Methods in Class

🔹 What is a Method?

A function inside a class.

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def display(self):
        print(self.name, self.marks)

s1 = Student("Alex", 75)
s1.display()
```

📌 Part 5: Real-World Example

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def is_pass(self):
        if self.marks >= 50:
            return "Pass"
        else:
            return "Fail"

s1 = Student("John", 40)
print(s1.is_pass())
```

📌 Part 6: Multiple Objects

```
students = [
    Student("A", 70),
    Student("B", 90),
    Student("C", 40)
]

for s in students:
    print(s.name, s.is_pass())
```

📌 Part 7: Real-World Thinking

👉 In AI projects:

- Data = objects
- Models = classes
- Functions = methods

👉 Example:

- Student → object
- Dataset → list of objects

🎯 Tasks for Day 10

🧠 **Task 1: Theory**

1. What is OOP?
2. Difference between class and object
3. What is constructor?
4. What is method?
5. Why OOP is used?

💻 **Task 2: Class Practice**

1. Create class Car with:
    - name
    - price
2. Create object and print values

💻 **Task 3: Constructor Practice**

1. Create class Student
2. Initialize name and marks
3. Print details

💻 **Task 4: Method Practice**

1. Create method to:
    - Check pass/fail
    - Calculate grade

💻 **Task 5: Object List**

1. Create list of students
2. Print all students
3. Find topper

🔥 **Task 7: Logic Challenge**

1. Create class BankAccount
    - deposit
    - withdraw
    - check balance
2. Create class Product
    - name, price
    - calculate discount